SHIVAM GUPTA

AI & Machine Learning Engineer

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TECHNICAL SKILLS

Languages: Python, C, JavaScript, TypeScript, HTML, CSS | **Libraries & Frameworks: R**eact.js, Next.js, Express.js, OpenCV, MediaPipe, NumPy, Tailwind CSS | **AI & Machine Learning:** Machine Learning, Deep Learning, Computer Vision, LLMs, RAG, Prompt Engineering, AI Agents, Pose Estimation, Semantic Search
Tools: Git, GitHub, VS Code, Postman, Power BI, FFmpeg, yt-dlp | **Databases & Cloud:** Elasticsearch, Elastic Cloud | **Concepts:** Data Structures & Algorithms (DSA), REST APIs, Full-Stack Development, Agentic AI, AI Automation, Real-Time AI Systems

PROJECTS

FITVEDA – AI Fitness & Dance Coaching

- Developed a **real-time AI Dance & Fitness Coach** that analyzes user movements from **YouTube dance and workout videos** using **MediaPipe Pose**, delivering **live pose scoring** with **gesture-based controls**.
- Integrated **OpenAI GPT-4o-mini**, **Groq Vision**, and **Text-to-Speech (TTS)** to provide **real-time AI voice coaching**, **performance evaluation**, and **personalized feedback**.
- Built an **end-to-end web application** using **Next.js, React, TypeScript, Tailwind CSS, and Modal (SAM2)**, featuring **AI-generated workout videos** and a **Spotify Wrapped-style performance report** with **personalized insights**.

Delayed AI Clone (Computer Vision)

- Developed an **AI-powered Delayed Clone System** using **Python, OpenCV, and MediaPipe** to create a **ghost-like delayed clone** while tracking **real-time body, hand, and face landmarks**.
- Implemented **pose, hand, and face mesh detection** with **temporal frame buffering** to generate a **smooth delayed clone effect** while keeping the live user unannotated.
- Optimized the **computer vision pipeline** for **low-latency, real-time performance** through efficient **landmark processing and rendering**.

F1 Replay System

- Developed an **interactive Formula 1 Race Replay application** using **Python, FastF1, Arcade, and NumPy** to visualize **real-time race telemetry** and **driver positions** on a rendered track.
- Implemented **real-time race analytics**, including **playback controls, live leaderboard, lap timing, tyre compounds, DRS status**, and **driver telemetry** (speed, gear, throttle, and brake).
- Built a **responsive GUI/CLI application** with **optimized telemetry processing and caching** to enable **smooth replay and analysis** of **Formula 1 race and qualifying sessions**.

EDUCATION

- **Lingayas Public School |** (Class X, CBSE) **| Percentage:80% |** 2021 - 2022
- **S.S. Khalsa. Sr. Sec. School |** (Class XII, CBSE) **| Percentage:70% |** 2023 - 2024
- **Aravali College of Engineering and Management**
  B.Tech. in Computer Science and Engineering (Specialization in Artificial and Machine Learning)
  2024 – 2028 (Pursuing)

CERTIFICATIONS

Microsoft Certified: Power BI Data Analyst Associate (PL-300) — Microsoft | 2026 | Credential

Microsoft Certified: Fabric Data Engineer Associate (DP-700) — Microsoft | 2026 | Credential

Complete Data Science, Machine Learning, Deep Learning & NLP Bootcamp

ACHIEVEMENTS

- **Winner of 1 National-Level Hackathon**
- **Top 10 Finalist among 1,000+ teams at the Chandigarh University Hackathon**
- **Qualified for India Innovates 2026, a nationwide innovation competition**
- **Participated in 10+ National-Level Hackathons, gaining experience in AI, full-stack development, and rapid prototyping**